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EAST-SPL

Event-Aware Statistical Tiling for Decomposable Soccer Player Localization with an Auxiliary Rejection Network



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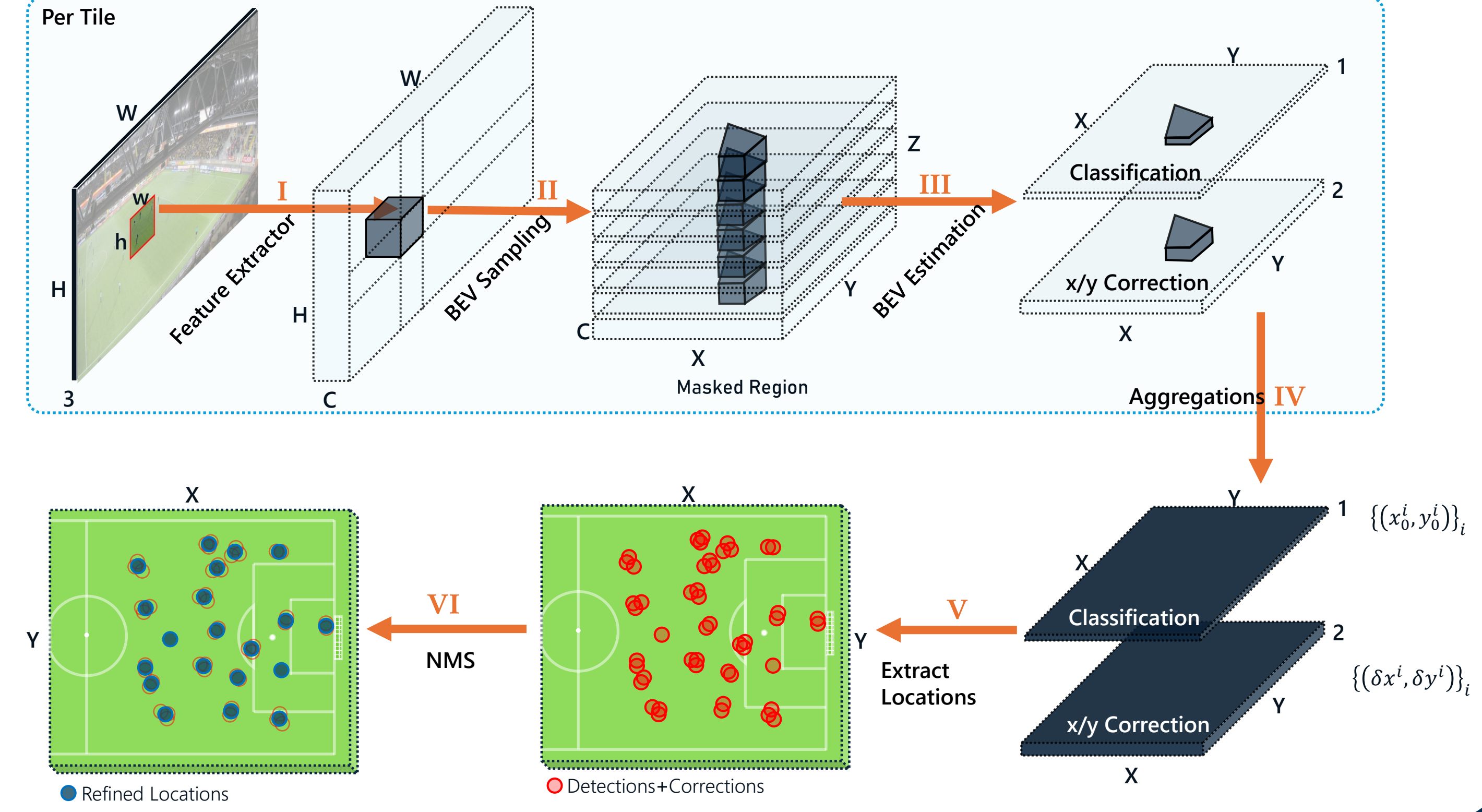
Abstract

In this work, we investigate the **statistically efficient** input-image tiling for **Soccer Player Localization**. In our previous work, **DTSP-*BEV*** [2], we introduced tiling to overcome memory limitations and enable parallel execution of **SPL-*BEV*** [3], a bird's-eye-view architecture for monocular soccer player localization. **EAST-SPL** extends this work through:

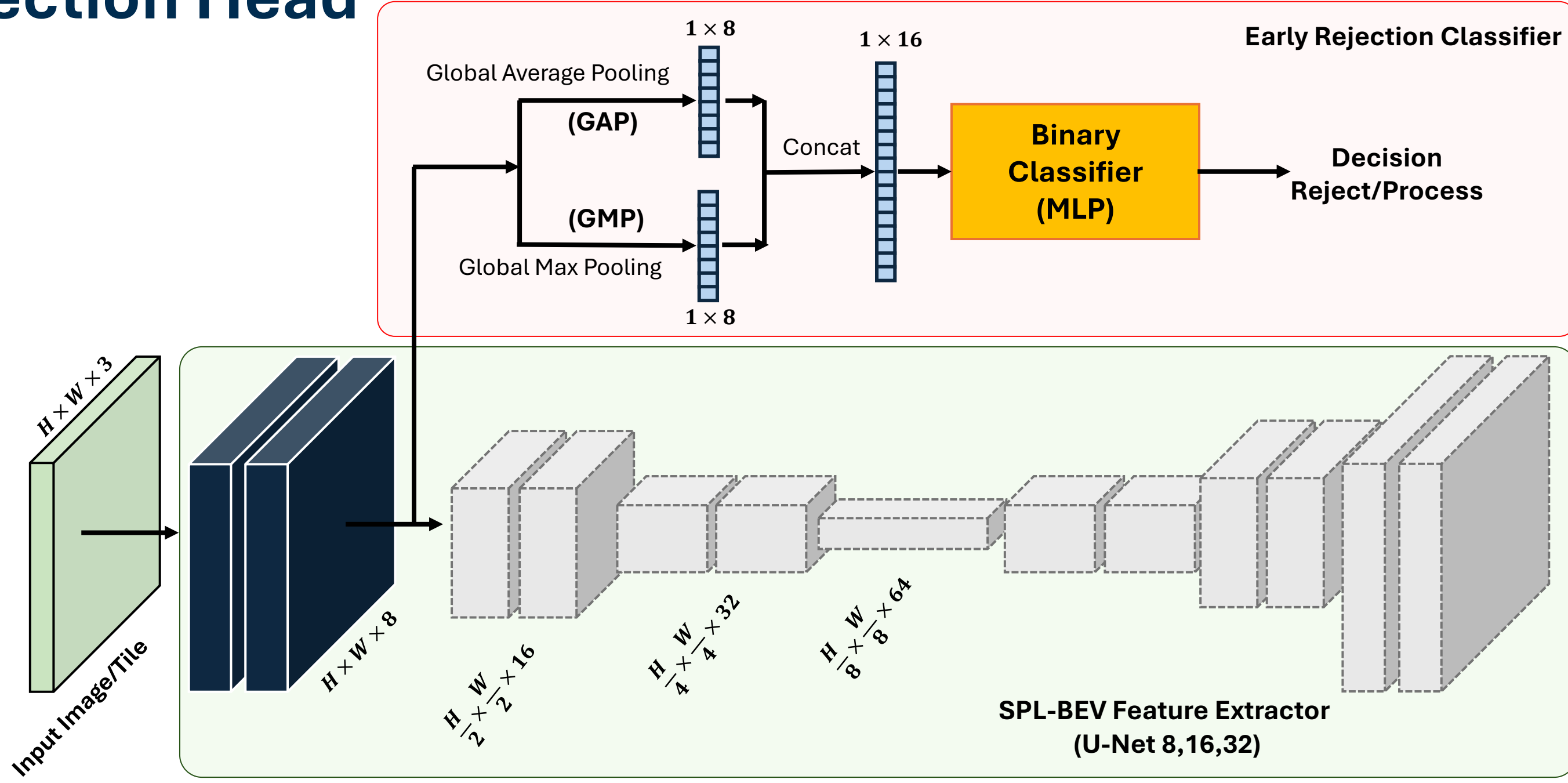
1. **Statistical formulation of objective function:** We minimize **Expected Total FLOPs ($\mathcal{ETF}$)** instead of the Total FLOPs of running the network on the whole image used in DTSP-*BEV* [2].
2. **An auxiliary rejection network** that bypasses empty tiles during inference, enabling statistically efficient tile configuration to reduce computation in practice.
3. **A new tile search optimization algorithm based on Genetic Algorithm (GA)**, which shows faster convergence than the Grid-Search method used in DTSP-*BEV* [2].

Background

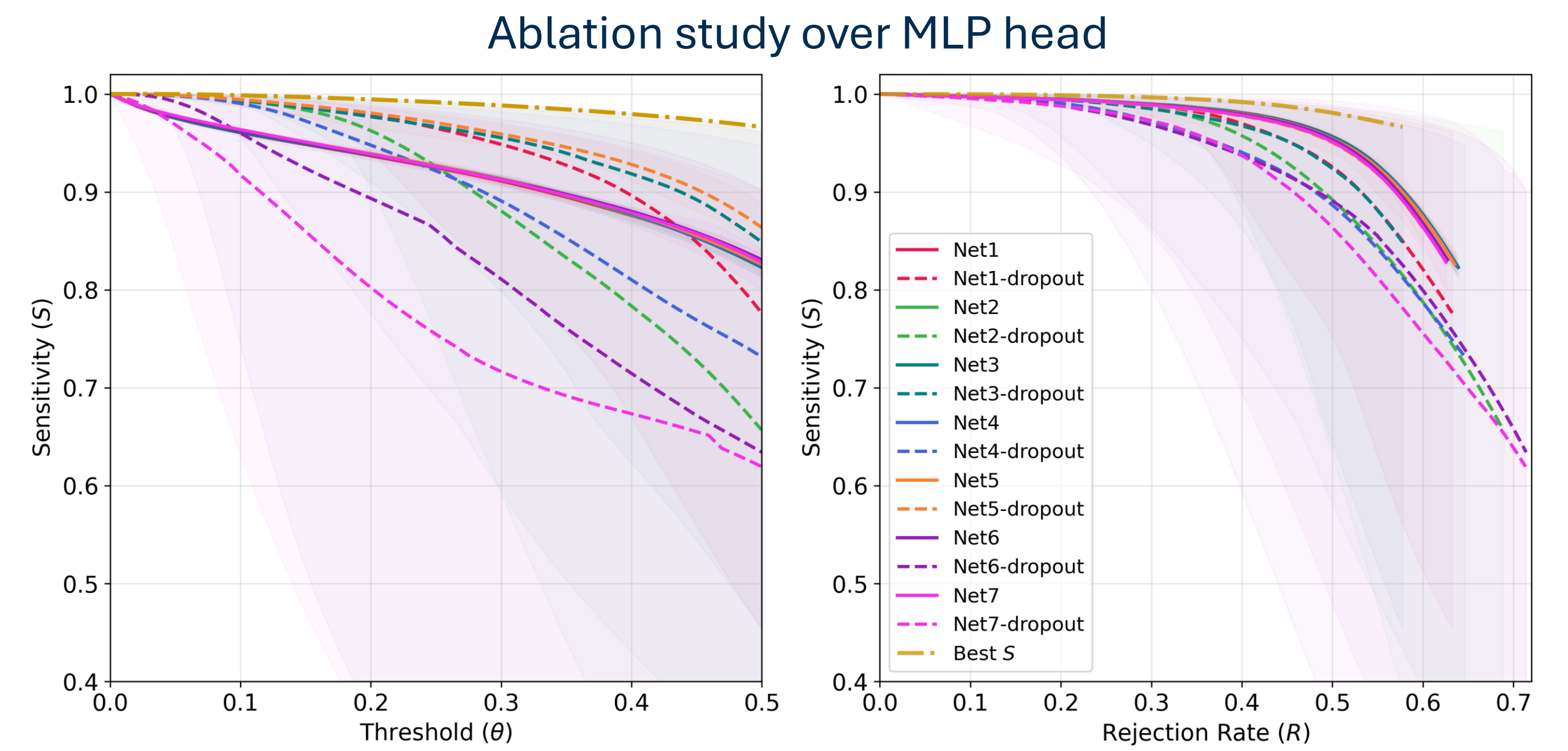
SPL-*BEV* & DTSP-*BEV*



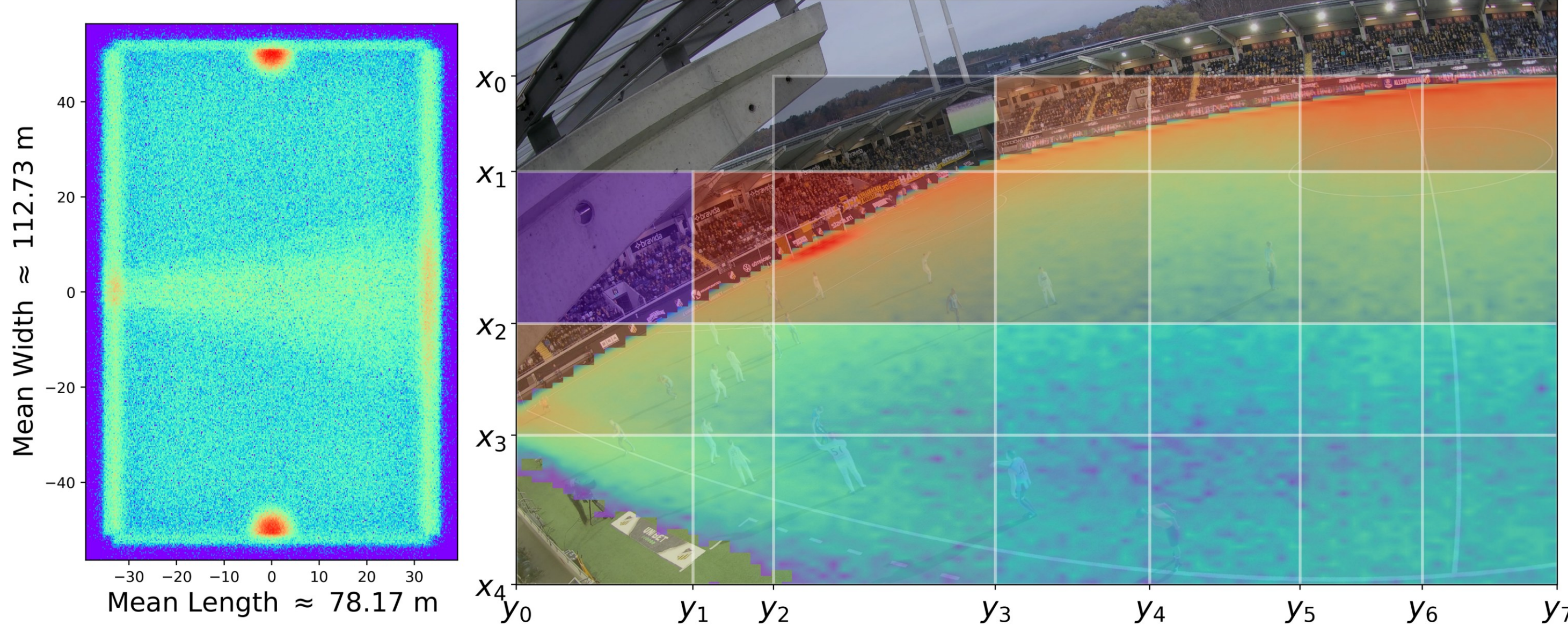
Rejection Head



Isolated training of Rejection Head



Statistical Formulation of Tile Configuration and $\mathcal{ETF}$



$$\mathcal{ETF}^{(\mathcal{M}; \mathcal{N})}(\mathbb{T}) := \sum_{t \in \mathbb{T}} \left(\mathcal{F}^{(\mathcal{O} + \mathcal{N})}(t) + \underbrace{\mathbb{P}[N(\text{Frame}[t]) = 1]}_{\mathbb{P}[\text{player} \subset t]} \cdot \left(\mathcal{F}^{(\mathcal{M})}(t) - \mathcal{F}^{(\mathcal{O})}(t) \right) \right)$$

$\mathcal{M}$: Primary network

$\mathcal{N}$: Rejection Head

$\mathcal{O}$: Shared Block

$\mathcal{F}^{(\mathcal{N})}(t)$: FLOPs of running network $\mathcal{N}$ on tile t

$\mathbb{T}$: Tile configuration, i.e., set of tiles

Tile Optimization Problem

Setup:

Collocated points

Soccer Field Mask

Projected Statistics to approximate

For a given $\mathcal{P}$, we can compute $\mathcal{ETF}$:

$$\mathbb{T}^* = \arg \min_{\mathbb{T}} \mathbb{E}_{\tau} [\mathcal{F}^{(\mathcal{M}; \mathcal{N})}(\mathbb{T}, \tau)]$$

$$\mathcal{P} = \{\{x_i\}, \{y_j\}\}$$

$$\mathcal{M}$$

$$\mathbb{P}[\text{player} \subset t]$$

Tile Optimization via GA Results

Search	Objective	Pop. Size	Runtime (seconds)	$\mathcal{ETF}$ (GFLOPs)	$\mathcal{T}$ (GFLOPs)	$\mathcal{T} + \mathcal{M}$ (GFLOPs)
Grid Search	$\mathcal{ETF}$	-	96.7	38.8	-	308.2
	$\mathcal{ETF} + \mathcal{M}$	-	90.2	101.5	-	163.4
GA	$\mathcal{ETF} + \mathcal{M}$	50	2.3	47.9	195.4	162.9
		100	62.5	55.7	200.0	156.0
		100	54.2	43.9	217.2	158.2
	$\mathcal{ETF}$	50	52.0	24.1	232.5	165.8

Conclusion & Future work

EAST-SPL reduces the inference cost of SPL-*BEV* by combining an auxiliary rejection head with event-aware tile optimization. Rather than minimizing single-frame FLOPs, the proposed objective searches for tile configurations with lower **Expected Total FLOPs**, allowing empty or low-probability regions to be processed more efficiently.

Future work will evaluate the **end-to-end localization accuracy** with simultaneous training of the rejection head and conduct broader ablation studies on the main EAST-SPL hyperparameters.

References

- [1] Chaman Motlagh and Nilsson, "On the **Reproducibility** of the **EAST-SPL** Pipeline", RRPR at ICPR 2026.
- [2] Persson et al., "**DTSP-*BEV***: Decomposable Tiled Soccer Player Localization", ICPRAM 2026.
- [3] Persson et al., "**SPL-*BEV***: Soccer Player Localization and Birds-Eye-View Estimation", CAIP 2025.
- [4] Ardö et al., "**Spiideo SoccerNet SynLoc** - Single Frame World Coordinate Athlete Detection and Localization with Synthetic Data", VISAPP 2025.

Generalization of the EAST-SPL Pipeline Beyond Soccer Player Localization [1]

Code & Repository

